Why pointers??

- To return more than one value from a function
- To pass arrays &structure more conveniently to functions.
- Concise and efficient array access
- Access dynamic memory
- To create complex data structures
- To communicate information about memory.

- Array is collection of similar data elements in contiguous memory locations.
- Elements of array share the same name i.e. name of the array.
- And they are identified by unique index in the array called as subscript.
- Array indexing starts from 0.
- Checking array bounds should be done by programmer.

1D Array - Syntax

```
int main(){
  int arr[5] = { 1, 2, 3, 4, 5 };
  int index;
  for(index=0;index<5;index++)
   printf("\n%d",arr[index]);
  printf("size = %d",sizeof(arr));
}
```

1D Array - Fig

arr		0	1	2	3	4	index
0x100		1	2	3	4	5	data
0x100		0x100	0x104	0x108	0x112	0x116	address
		arr[0]	arr[1]	arr[2]	arr[3]	arr[4]	access

- **&arr[0] = 0x100**
- **&arr = 0x100**
- **arr = 0x100**
- ***arr = 1**

1D Array - Syntax

```
int main()
{
   int arr[5], index;

   printf("Enter 5 elements..");
    for(index=0;index<5;index++)
      scanf("%d",&arr[index]);

  for(index=0;index<5;index++)
      printf("\n%d",arr[index]);
  return 0;
}
```

1D Array & pointers - Syntax

```
int main() {
    int arr[5], *ptr, index;
    ptr = arr ;
    printf("Enter 5 elements..");
    for(index=0;index<5;index++)   {
        scanf("%d", ptr);
        ptr++;
    }
    ptr = arr;
    for(index=0;index<5; index++, ptr++)
         printf("\n%d",*ptr);
}
```

- **int arr[SIZE] ;**

arr[index] is internally resolved as *(arr+index)

arr[index] == *(arr+index)

***(index+arr) == index[arr]**

- **int *ptr=arr;**

***(ptr+index) == ptr[index]**

***(index+ptr) == index[ptr]**

Passing array to function

- **Arrays are passed to function by ref.**
- **The address of starting element of array (Base address) is passed to the function.**
- **The address is collected in a pointer.**
- **We can use pointer as well as array notation to access array elements in the function.**

Pointer to pointer

- **The pointer that stores address of another pointer variable is called as 'pointer to pointer'.**
- **Example :-**

```
int x=10;   int *px=&x;
int **ppx=&px;
printf("%d",**ppx);
```